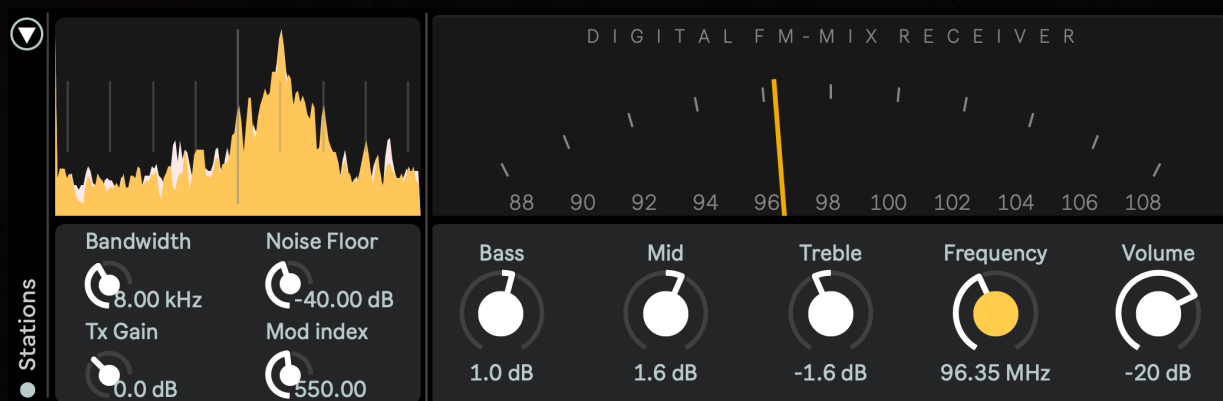
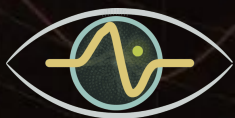




MAX FOR LIVE DEVICE

FM Radio

User Manual

Tune and mix up to eight stereo signals as stations on the same FM band, inside Ableton Live.

Installation

FM Radio is distributed as a single Max for Live device file, `FM_Radio.amxd`, from ssll.gumroad.com/l/fmradio. There is no installer and nothing to authorise.

- 1 Download the `.zip` from your Gumroad library and unpack it. Inside you will find the device file, `FM_Radio.amxd`, together with `FM_Radio_Manual.pdf` and `LICENSE.txt`.
- 2 Copy `FM_Radio.amxd` into your Ableton User Library. I recommend giving it a folder of its own, such as `Devices/SSLL`.
- 3 Open **User Library** in Live's browser and locate the FM Radio device in the folder where you placed it.

Requirements

FM Radio has been tested and verified with Ableton Live 12 Suite.

Version, updates and removal

The installed version is shown by hovering the circle below the `Stations` label. This manual covers version 1.0.0.

To update, replace the `.amxd` in your User Library with the new one. A device already placed in a saved set keeps the copy that was frozen into that set; open the set and drag in the new version to replace it.

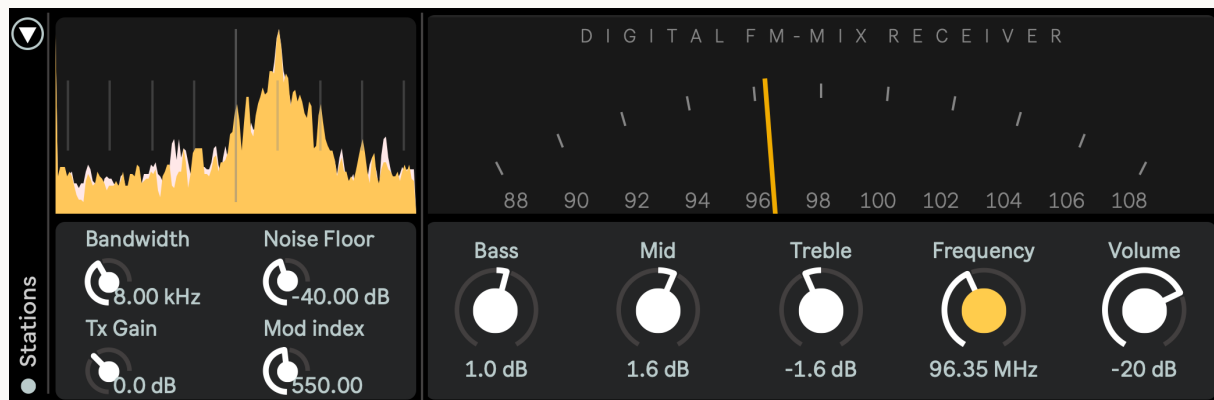
To uninstall, delete `FM_Radio.amxd` from the User Library. Nothing else is written anywhere on the system.

About the device

FM Radio brings the physical imperfections of broadcast radio into Ableton Live. Place up to eight stereo sources across a shared FM band and tune between them using a virtual receiver. Shape transmission bandwidth and noise, and adjust the receiver colour with a three-band EQ.

What separates it from an effect preset that imitates a bad radio signal is that nothing here is imitated. Each station is genuinely frequency-modulated onto its own carrier and the receiver genuinely demodulates whatever it finds at the frequency you have tuned to.

FM Radio is an audio effect. It runs on one track in Live (either a MIDI or an audio track) and pulls in the eight stations from elsewhere in the set based on the settings in the Stations panel — see [Audio routing](#).

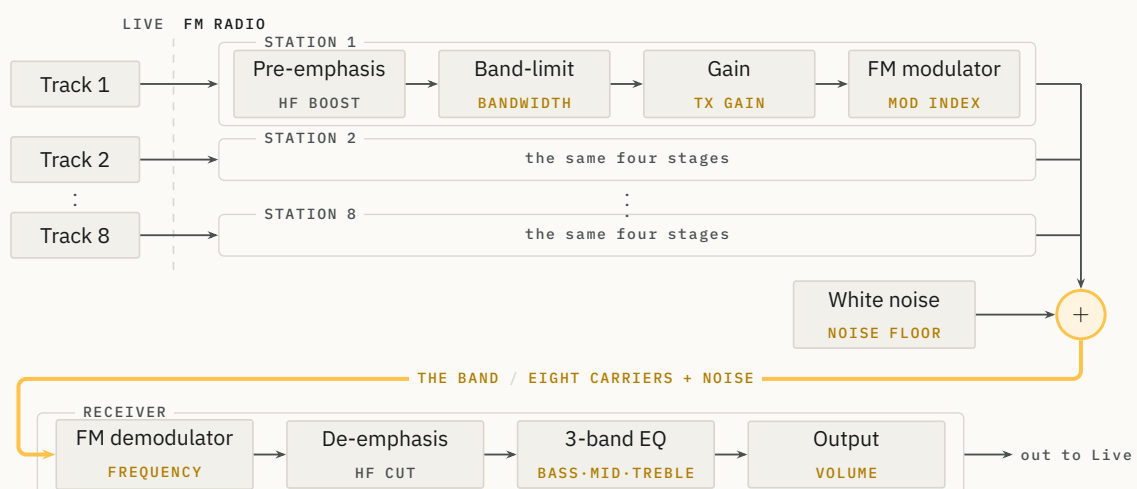


The device as it appears in Live, with the Stations panel collapsed. Left: the band spectrum and the transmitter controls. Right: the tuning dial, the receiver EQ, and the receiver output.

How it works

The signal path is a complete FM transmission chain, running at audio rate inside Live.

- 1 Transmit:** Each of the eight stations takes audio from a track in Live, pre-emphasises and band-limits it, and frequency-modulates it onto its own carrier.
- 2 Channel:** The eight modulated carriers are summed into one band and white noise is added.
- 3 Receive:** The receiver bandpass-filters the band around the frequency you have tuned to, demodulates what it finds, undoes the pre-emphasis, and passes the result through a three-band EQ.



FM Radio signal diagram: Eight stereo tracks in, one receiver stereo signal out. Every station runs the same four stages, and the band is the one thing they all share.

Pre-emphasis and de-emphasis are the pair a real broadcast chain uses. The transmitter lifts the high end before modulating, the receiver takes the same amount back out afterwards, and the hiss the channel added in between comes down with it.

Because the receiver is tuned continuously rather than switched between stations, everything that happens between two carriers happens for real: partial capture, the two stations beating against each other, the noise floor rising as you leave a carrier behind.

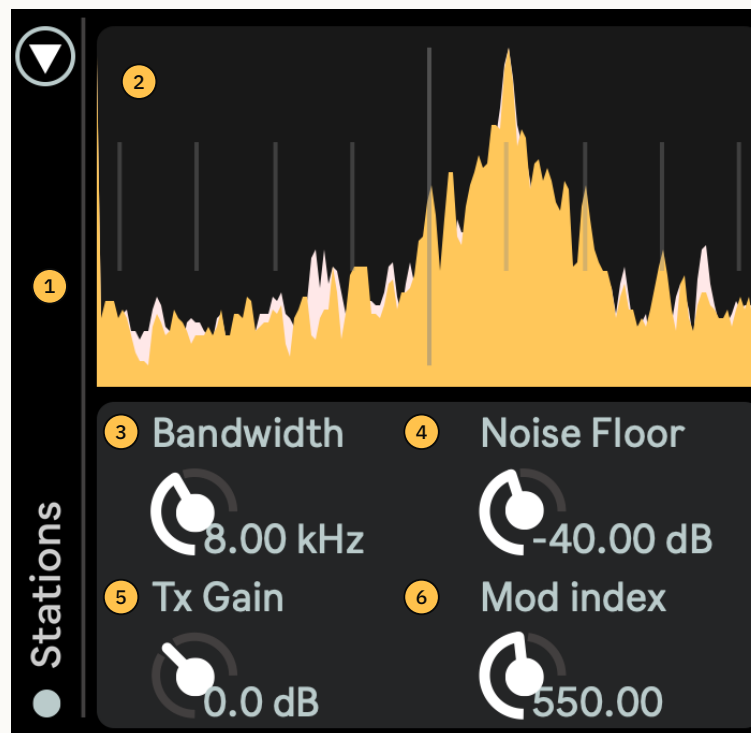
NOTE

Since the band inside FM Radio is audio-rate, the host sample rate sets how much room there is in it: the higher the sample rate, the more FM bandwidth is available. Running Ableton at 48 kHz is recommended.

Parameters and audio routing

The panel divides in two along the middle rule. Left of it are the FM modulation settings and a spectrum display of the received FM band. Right of it are the *receiver* controls: the tuning dial, the demodulation frequency, and the EQ and volume. On the top left is the arrow that opens the Stations panel, covered under [Audio routing](#).

Transmitter and band



- ① Stations — opens and closes the Stations panel.
- ② Band spectrum — the whole band as the receiver sees it.
- ③ Bandwidth, ④ Noise Floor, ⑤ Tx Gain, ⑥ Mod index — the transmitter controls.

The amber spectrum is the summed band: the eight modulated carriers plus the noise floor. The central grey line marks the centre frequency demodulated by the receiver. The cleanest demodulated signal is obtained when the transmit carrier and the demodulation frequency match, which happens when the received spectrum peak aligns with the centre line.

Bandwidth

The cutoff of the lowpass filter applied to every station *before* modulation. This is the transmitted audio bandwidth, and the single biggest contributor to the lo-fi character. Low settings narrow every station towards a telephone-like band; high settings keep the source closer to full range, but each station then occupies more of the shared band and interferes with its neighbours sooner.

Noise Floor

The level of the white noise added to the band, after the carriers are summed and before the receiver gets to it. It belongs to the channel rather than to any one station, so it sets how far a station can be from the tuned frequency before it disappears and how much hiss sits underneath one you are tuned straight to.

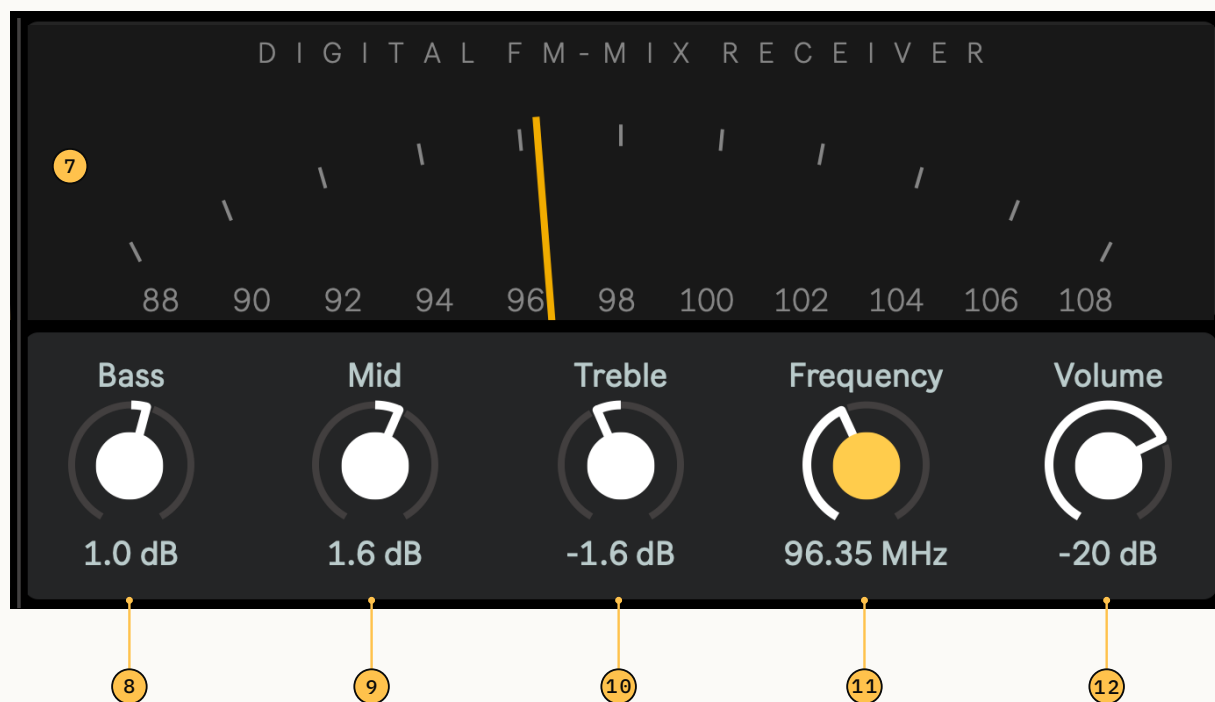
Tx Gain

Input gain into the modulator, applied to every station equally. The modulator clips the frequency deviation at its limit, so this is a drive control: past a certain point the extra level stops making the station louder and starts making it distort, the way an over-deviated transmitter does.

Mod index

The frequency deviation — how far each station's carrier is pushed from its centre by the audio driving it, and so how much of the band a single station occupies. Low values keep stations narrow and well separated, with silent gaps between them. High values widen every station until they overlap and the band turns into stations bleeding through one another.

Receiver



7 Tuning dial — the band, and where the receiver sits on it.

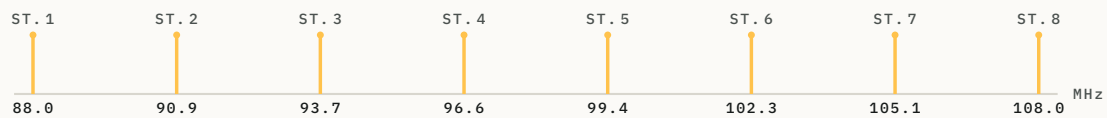
8 Bass, 9 Mid, 10 Treble, 11 Frequency, 12 Volume — the receiver controls.

Frequency

RANGE 88.00 - 108.00 MHz

Where the receiver is tuned. Tune slowly across the stations and you will hear them rise out of the noise, lock, and then distort as you pass through towards the next station.

The eight carriers are spaced evenly across the dial, 2.86 MHz apart, with `st. 1` sitting at the bottom of the band and `st. 8` at the top, so tuning from one end to the other passes through all eight in order:



Note that the scale of 88.00 – 108.00 MHz is an arbitrary design choice to mimic a real radio's dial. Carrying a real FM band would need a sample rate in the hundreds of megahertz. Instead every carrier in FM Radio sits inside the audio band, at a few kilohertz, where Live's sample rate can carry it. This is also why the host sample rate sets how much room the band has to work with.

Bass / Mid / Treble

A three-band EQ on the receiver output, after demodulation. These are the receiver's colour adjustments. Because it sits after demodulation it shapes the noise and the distortion along with the demodulated audio.

Volume

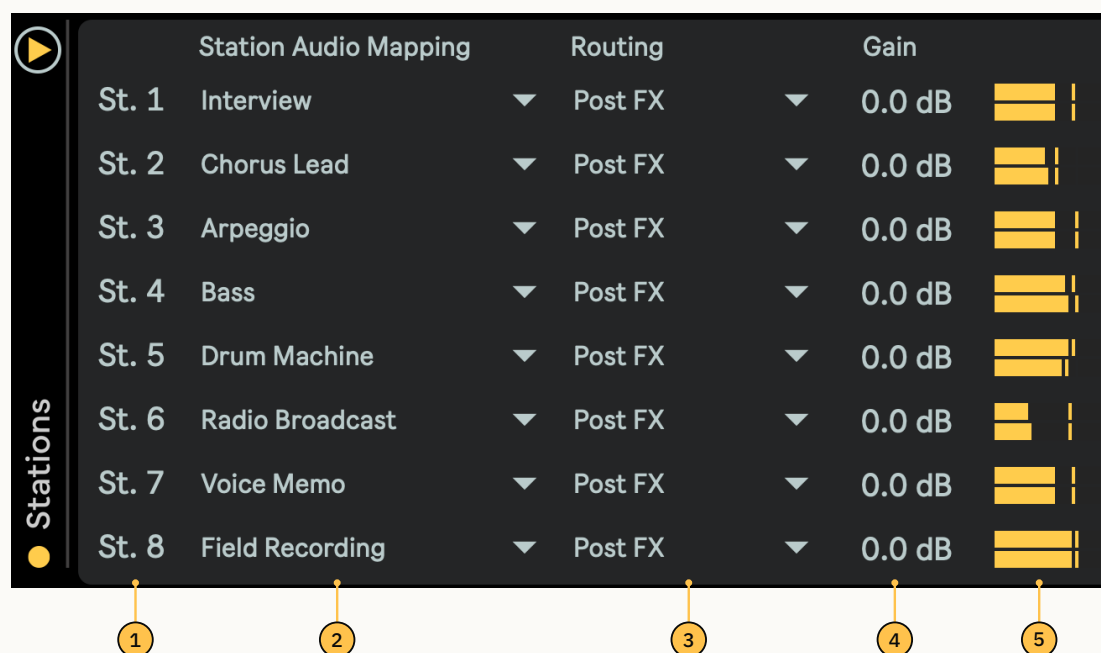
Output level of the device, applied last. It defaults well below unity, because a demodulated band with the noise floor up can get loud quickly and the sum of eight stations is not a predictable level.

TIP

Every control is automatable and mappable from Live in the usual way. **Frequency** in particular is worth mapping to a knob, LFO, or automating across a section in the Live arrangement view.

Audio routing

FM Radio sits on one track but listens to eight. Click `stations` on the left edge of the panel to fold the Stations list out, and each of the eight rows becomes a transmitter you can point at any track in the set.



The Stations panel, folded out to the left of the transmitter controls.

- ① St. 1–8 — the eight stations, in band order.
- ② Station Audio Mapping — which track this station transmits.
- ③ Routing — the point in that track's chain the audio is taken from.
- ④ Gain — trim for this station alone, before the transmitter.
- ⑤ Level meter — what is actually arriving.

To set a station up: choose its source with **Station Audio Mapping**, which lists every track in the set; choose where in that track the audio is tapped with **Routing**; and use **Gain**, with the meter beside it, to get the eight stations arriving at comparable levels. Then set **Frequency** to that station's carrier and it comes out of the noise.

ROUTING POINTS

SETTING	TAKES THE AUDIO
Pre FX	Straight from the clip or input, before the track's own devices. The station transmits the raw source no matter what is on the track.
Post FX	After the track's devices, before its volume fader and pan. The track's own processing is transmitted; its fader is not. Default.
Post Mixer	After the fader and pan as well, so the track's mix moves affect what is transmitted.

NOTE

The routing is a tap, not a move. Choosing a track as a station does not take it out of the mix — it keeps playing through its own output as usual, and the station transmits a copy. If you want to hear a track *only* through the radio, turn that track's monitoring or output down and leave FM Radio to carry it.

Copyright and contact

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The open-source core

The DSP core began as *radiofm*, a set of Max/MSP abstractions for FM modulation, demodulation and channel simulation. Those abstractions are published separately as open source at github.com/hefo/maxmsp and are free to study, adapt and build on. The implementation is described in:

H. Forssell, “[Simulating FM Radio Transmissions in Max/MSP](#)”, *PdMaxCon25*~, University of Illinois Urbana-Champaign, September 2025.

ssll.fm/publications

The separately published *radiofm* materials remain available under their own open-source licence. The commercial licence for FM Radio does not replace or restrict the rights granted under that licence.

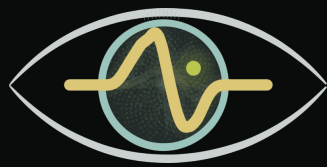
No warranty

FM Radio is provided “as is”, without warranty of any kind, express or implied, to the fullest extent permitted by applicable law. To the fullest extent permitted by applicable law, Henrik Forssell is not liable for loss or damage arising from its use, including lost data, lost work, or damage caused by unexpected output levels.

Get in touch

If something does not work, or you want to share thoughts and ideas or what you made with the device, write to ssll.musik@gmail.com.

When reporting a problem, it helps to include your Live version, your Max for Live version, your operating system, and your sample rate and buffer size.



FM Radio

Designed and built in Stockholm by Henrik Forssell.

Version 1.0.0 / September 2026

SSL.FM